Full Name:	Reg. No.:

MZUZU UNIVERSITY

Mathematics Department

EMT 1101 - Precalculus Continuous Exam II

September 17, 2012

Time Allowed : 2 hours

Instructions

- Check that the paper contains **8 questions**
- Attempt **all** questions in the spaces provided on the question sheets

1. (a) Use the quadratic formula to show that the roots of the equation $x^2 + 3x + 1 = 0$ are $\frac{1}{2}\left(-3 \pm \sqrt{5}\right)$.

3 marks

(b) Show that $\frac{1}{\frac{1}{2}(-3-\sqrt{5})} = \frac{1}{2}\left(-3 + \sqrt{5}\right)$.

3 marks

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2. Solve for x

(a) $\mid 3x + 5 \mid > 17$.

3 marks

(b) $\left|\frac{x+1}{2} - \frac{x-1}{3}\right| < 1$

3 marks

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3. (a) If $g(x) = x^2 - 3xk - 4$ and $g(1) = -2$, find k.

3 marks

(b) Let $f(x) = \sqrt{\frac{4x-3}{x^2-4}}$. Find the domain of $f(x)$

8 marks

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4. Given that $f(x) = \dfrac{-4x}{6x+1}$ and $g(x) = \dfrac{1-5x}{3x}$, compute the following.

(a) $f^{-1}(x)$

3 marks

(b) $g^{-1}(x)$

3 marks

(c) $f^{-1}(g^{-1}(x))$.

3 marks

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5. (a) Find the equation of the line passing through $(-3, 0)$ that is perpendicular to the straight line $2x + y - 2 = 0$.

5 marks

(b) A line joins the points $(2, 3)$ and $(5, 1)$.

i. Sketch the line;

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ii. Sketch the reflection of the line about the x-axis;

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iii. What is the equation of the line of symmetry.

1/1/1 marks

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6. (a) Let $f(x) = 3x^2 - 4x$ and $g(x) = 2x + 1$. Compute each of the following:

i. $(f - g)(x)$

2 marks

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ii. $f(g(-1))$

2 marks

(b) Let $g(x) = 4x-1$. Find $f(x)$, given that the equation $(g \circ f)(x) = x+5$ is true for all values of x.

4 marks

(c) Define a function f, with domain the positive integers, as follows:

$$f(x) = \begin{cases} 3x + 1 & \text{if } x \text{ is odd} \\ \frac{x}{2} & \text{if } x \text{ is even} \end{cases}$$

Compute the first three iterates of $x_0 = 1$.

3 marks

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7. (a) If $f(x) = \frac{-3}{x^2}$, find and simplify the difference quotient $\dfrac{f(x+h) - f(x)}{h}$ for the function.

5 marks

(b) Let $f(x) = \sqrt{x}$. Find a number b so that the average rate of change of f on the interval $[1, b]$ is $\dfrac{1}{7}$.

6 marks

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8. (a) Solve $x^{\frac{2}{3}} - x^{\frac{1}{3}} = 1$

7 marks

(b) If the roots of the equation $4x^2 - 6x + 1 = 0$ are r_1 and r_2, find

i. the value of $r_1^3 + r_2^3$,

5 marks

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ii. the equation whose roots are $\left(r_1 + \frac{1}{r_2}\right)$ and $\left(r_2 + \frac{1}{r_1}\right)$

6 marks

(You may wish to use the following identities:

i. $r_1^3 + r_2^3 = (r_1 + r_2)(r_1^2 - r_1 r_2 + r_2^2) = (r_1 + r_2)\left[(r_1 + r_2)^2 - 3r_1 r_2\right]$

ii. $\left(r_1 + \frac{1}{r_2}\right) + \left(r_2 + \frac{1}{r_1}\right) = (r_1 + r_2) + \frac{r_1 + r_2}{r_1 r_2}$

iii. $\left(r_1 + \frac{1}{r_2}\right)\left(r_2 + \frac{1}{r_1}\right) = r_1 r_2 + 2 + \frac{1}{r_1 r_2}$)

`End of Questions`